

MarketPay Testing Guide

MarketPay — Testing Guide & System Overview

Last updated: 22 June 2026

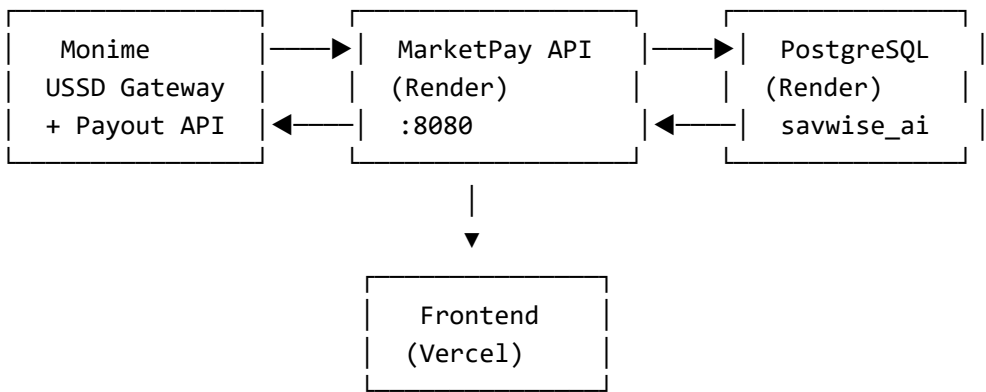
Base URL (prod): <https://market-pay-fpsi.onrender.com>

Frontend: <https://market-pay.vercel.app> (manual redeploy required after backend push)

Table of Contents

- 1. [System Architecture](#)
- 2. [USSD Flows](#)
- 3. [Vendor Registration \(New\)](#)
- 4. [Loan Disbursement](#)
- 5. [Repayment Tracking \(New\)](#)
- 6. [Test Accounts](#)
- 7. [Testing Scenarios](#)
- 8. [Key Decisions](#)

1. System Architecture



Key Components

Component	Technology	Location
API Server	Go + Gin + GORM	cmd/api/main.go
Database	PostgreSQL	Render (savwise_ai)
Migrations	SQL files	migrations/000001-000011
USSD Exchange	Monime platform + RSA crypto	internal/monime/exchange/
Payouts	Monime Payout API	pkg/monimepayout/
Webhooks	Monime v2 payload	internal/monime/interfaces/http/webhook.go
Frontend	Next.js (standalone)	market_pay-frontend/

2. USSD Flows

2a. Public Flow (monime-ussd-public-flow.json)

No access gate — any mobile subscriber can use it.

Welcome to MarketPay

1. Pay a vendor
2. Register as vendor
3. Exit

Pay a Vendor Path

- Enter vendor code (e.g. MP12345)
- Enter amount in SLE (min 1)
- Confirm & pay / Cancel
- PIN entry (Monime platform handles collection)
- Result displayed

Register as Vendor Path

- Enter full name
- Enter market name
- Enter phone number (e.g. 23276123456)
- Confirm / Cancel
- Result: "Registration submitted. A field agent will contact you."

2b. Vendor Flow (monime-ussd-vendor-flow.json)

Access gate required — subscriber must be in ussd_allowed_subscribers.

- Access Gate (SHA-256 subscriberId check)
 - Menu:
 1. My credit score
 2. Apply for loan
 3. Repay loan
 4. Check balance
 5. Transaction history
 6. Exit

Repay Loan Path

- Shows outstanding balance
- Enter repayment amount
- Confirm & pay / Cancel
- PIN entry (Monime platform handles collection)
- Result displayed
- LoanRepayment record created (PENDING)
- Webhook financial_account.credited → confirmed (COMPLETED)

3. Vendor Registration

Flow Steps

1. Subscriber selects “Register as vendor” on public USSD menu
2. Collects: name → market → phone number
3. Phone normalization: normalizePhone() handles all formats:
 - 23276123456 → +23276123456
 - 076123456 → +232076123456
 - +23276123456 → +23276123456
4. Creates users record with is_active = false
5. Creates vendors record with Status = PENDING
6. Links subscriber to vendor in ussd_subscribers
7. NOT added to ussd_allowed_subscribers yet

Approval (Waiting Room)

After USSD registration, the vendor is in a “waiting room”:

State	Can log in?	Can use vendor USSD menu?
Just registered (PENDING)	No (is_active = false)	No (not in allowed_subscribers)
Field agent approves KYC	Yes (is_active = true)	Yes (if subscriberId added)

Approval endpoint: PUT /api/v1/vendors/:id/kyc/approve

- Sets vendor status → ACTIVE - Sets KYC status → VERIFIED - Sets users.is_active = true - To also grant USSD access, add subscriberId to ussd_allowed_subscribers (via DB or future endpoint)

Registration Message

“Registration submitted for [Name] at [Market]. Phone: [Phone]. A field agent will contact you to verify and activate your account.”

4. Loan Disbursement

States

APPROVED → DISBURSEMENT_PENDING → ACTIVE (on payout.comPLETED)
→ APPROVED (on payout.failed)

Disburse Action

PUT /api/v1/loans/:id/disburse

1. Validates loan is APPROVED and user has permission
2. Determines mobile money provider from phone prefix:
 - +23276/77/78 → m17 (Orange SL)
 - +23230/33/34 → m18 (Africell)
 - configurable via provider_mappings in config.yaml
3. Calls Monime Payout API (POST /v1/payouts)
4. Sets loan state to DISBURSEMENT_PENDING
5. Stores monime_reference = payout ID

Payout Webhooks

Event	Handler	Effect
payout.comPLETED	handlePayoutCompleted	State → ACTIVE, generates schedules, sets provider_ref
payout.failed	handlePayoutFailed	State → APPROVED, clears ref, stores failure_reason

Failed Payout — Retry

If a payout fails, the loan returns to APPROVED. The three-dot menu on the loans page allows re-disbursing.

5. Repayment Tracking

Table: loan_repayments

Column	Type	Purpose
id	UUID (PK)	Primary key
loan_id	UUID (FK→loans)	The loan being repaid
vendor_id	UUID (FK→vendors)	The vendor repaying
amount	DECIMAL(15,2)	Repayment amount
monime_ref	VARCHAR(255)	Monime collection reference (webhook match key)
payment_ref	VARCHAR(255) UNIQUE	Our internal reference (REPAY-{session}-{ts})
metadata	JSONB	Session data, masked_phone, subscriber_id
status	VARCHAR(50)	PENDING → COMPLETED / FAILED
paid_at	TIMESTAMPTZ	When confirmed
created_at / updated_at	TIMESTAMPTZ	Audit timestamps

How Repayments Work

USSD Repayment Flow:

1. Vendor selects "Repay loan" → sees balance
2. Enters amount → confirms
3. Monime collect_payment template processes PIN & collection
4. Callback hits handleRepaymentResult:
 - Extracts ExternalRef from ExportedData (or falls back to payment_ref)
 - Creates LoanRepayment record (status=PENDING)
 - Shows result to user
5. Monime sends webhook financial_account.credited:
 - handleAccountCredited matches by monime_ref (object.id)
 - Fallback: matches by metadata.payment_ref
 - Calls RepaymentService.ConfirmRepayment()
 - marks LoanRepayment as COMPLETED
 - calls existing Repay() to apply to loan schedules

Webhook Matching Strategy

1. **Primary:** Match financial_account.credited.object.id against loan_repayments.monime_ref
2. **Fallback:** Check data.metadata.payment_ref against loan_repayments.payment_ref

If no match is found, the repayment stays PENDING and must be reconciled manually.

6. Test Accounts

Vendors (Login via phone + PIN on website)

Name	Phone	PIN	Mobile Money
Araia	+23276902971	2006	Orange SL (m17)
Moses Moore	+23233346989	2004	Africell (m18)
Nelson	+23279826564	2005	Orange SL (m17)
Bernard (Gamanga)	+23272571067	2005	Orange SL (m17)

Loan Officer (Login via email + password)

Name	Email	Password	Role
Joshua Yoki	joshua@marketpay.sl	password123	LOAN_OFFICER

Vendor Phones by Network

Network	Prefix	Monime Provider ID
Orange SL	+23276, +23277, +23278	m17
Africell	+23230, +23233, +23234	m18

7. Testing Scenarios

Scenario 1: Public Pay Vendor

1. Dial USSD code (public flow)
2. Select “Pay a vendor”
3. Enter vendor code MP12345 (or any valid code)
4. Enter amount (e.g. 10)
5. Confirm → PIN entry → success
6. **Verify:** Notification sent to LOAN_OFFICER role

Scenario 2: Public Vendor Registration

1. Dial USSD code (public flow)
2. Select “Register as vendor”
3. Enter name → market → phone
4. Confirm
5. **Verify:** Message says “Registration submitted. A field agent will contact you.”
6. **Verify:** New vendor created with Status = PENDING, users.is_active = false

Scenario 3: Field Agent Approval

1. Log into website as LOAN_OFFICER
2. Navigate to vendor management
3. Find the new vendor
4. Click “Approve KYC”
5. **Verify:** Vendor status → ACTIVE, user → is_active = true
6. **Optional:** Add subscriberId to ussd_allowed_subscribers for USSD access

Scenario 4: Full Loan Cycle (Moses Moore — Africell)

Should succeed (Africell + m18):

1. Log in as LOAN_OFFICER
2. Create loan for Moses Moore (+23233346989)
3. Approve loan
4. Disburse loan (should use m18 provider)
5. **Verify:** Loan state → DISBURSEMENT_PENDING
6. Wait for webhook payout.completed
7. **Verify:** Loan state → ACTIVE, schedules generated
8. Moses checks balance via USSD → shows outstanding

Scenario 5: Full Loan Cycle (Araia — Orange SL)

Should succeed (Orange + m17):

1. Log in as LOAN_OFFICER
2. Create loan for Araia (+23276902971)
3. Approve loan
4. Disburse loan (should use m17 provider)
5. **Verify:** Loan state → DISBURSEMENT_PENDING
6. Wait for webhook payout.completed
7. **Verify:** Loan state → ACTIVE, schedules generated

Scenario 6: Repay via USSD (Vendor Flow)

Prerequisite: Vendor has an ACTIVE loan.

1. Dial USSD code (vendor flow)
2. Select “Repay loan”
3. Enter amount
4. Confirm → PIN entry
5. **Verify:** LoanRepayment record created (status = PENDING)
6. Wait for webhook financial_account.credited
7. **Verify:** LoanRepayment status → COMPLETED
8. **Verify:** Loan schedule updated (AmountPaid increased)

Scenario 7: Payout Failure → Retry

1. Disburse loan (any vendor)
2. Webhook returns payout.failed

3. **Verify:** Loan state → APPROVED, failure_reason set
 4. On loans page, click three-dot menu → “Retry Disbursement”
 5. **Verify:** New payout sent, state → DISBURSEMENT_PENDING again
-

8. Key Decisions

Webhook Signature

Monime sends empty X-Monime-Signature. We log a warning but accept the webhook. No 401 rejection.

Phone Normalization

normalizePhone() strips non-digits, handles 232..., 076..., +... formats. Outputs +232... format.

Subscriber Identity

- global.subscriberId = SHA-256 hash of MSISDN — used for identification
- global.subscriberMsisdn = masked (e.g. 233XX XXX 4567) — display only, never used as lookup

Provider Selection

Provider ID (m17 / m18) resolved from provider_mappings in config.yaml by phone prefix, not hardcoded.

Migration Auto-Apply

Render runs ./migrate up && ./app on startup. Migrations 000001–000011 apply automatically.

Loan Source

- WEB (default) — created via website
- USSD — created via USSD application flow